

INTRODUCTION

The purpose of this Industrial Training – I was to bridge the gap between theoretical knowledge and practical implementation. In college, we learn many concepts, but this training helped me understand how these concepts are actually used in real-life situations.

This training aimed to provide hands-on experience in programming using Core Java. It helped me improve my coding skills and understand how to write clean and efficient programs. I also learned how different concepts of Object-Oriented Programming (OOP) like classes, objects, inheritance, and polymorphism are applied in real software development.

Another important purpose of this training was to develop problem-solving ability and logical thinking. By working on practical tasks and small programs, I gained confidence in coding and debugging errors.

Overall, this training helped me understand the basics of software development, improved my technical knowledge, and prepared me for future academic projects and industry work.

Objectives

The main objectives of the training were:

- To understand the fundamentals of **Java Programming Language**
- To learn and implement **Object-Oriented Programming (OOP) concepts**
- To gain practical knowledge of **Core Java features** such as classes, objects, inheritance, and polymorphism
- To develop problem-solving and coding skills
- To gain exposure to real-time project development
- To enhance technical and logical thinking abilities

Overview of Training Program

The Industrial Training – I was conducted at **C-DAC, Patna** for a duration of two weeks, from **22nd December 2025 to 3rd January 2026**. The training sessions were held daily from 9:00 AM to 1:30 PM.

The training focused on **“OOP’s Concepts using Core Java”** and included both theory and practical sessions. In the beginning, basic concepts of Java were taught, such as data types, operators, and control statements. After that, we learned important Object-Oriented Programming (OOP) concepts like classes, objects, inheritance, polymorphism, and encapsulation.

As the training progressed, more advanced topics were covered, including interfaces, exception handling, file handling, collections framework, and multithreading. Each topic was explained with examples and practical coding, which made it easier to understand.

The sessions were conducted by experienced faculty members, **Mr. Vaibhav Patle** and **Ms. Nishu**, who guided us throughout the training and helped us clear our doubts. At the end of the training, we worked on a small project and gave a final assessment to test our understanding.

Overall, the training program was well-structured and helped me gain both theoretical knowledge and practical experience in Java programming.

COMPANY PROFILE

Introduction to the Organization

The **Centre for Development of Advanced Computing (C-DAC)** is a well-known research and development organization in India. It works under the Ministry of Electronics and Information Technology (MeitY), Government of India.

C-DAC was established to develop advanced technologies in the field of computing. Over the years, it has played an important role in the growth of Information Technology in India. It is especially known for its work in supercomputing, software development, and technical education.

C-DAC has many centers across India, and each center focuses on research, development, and training. The Patna center mainly focuses on providing training programs and skill development for students.

Location

The training was conducted at **C-DAC, Patna**, located in Bihar, India. This center provides a good learning environment with proper labs, systems, and experienced faculty members.

Services and Areas of Work

C-DAC works in many advanced areas of technology. Some of the main services and fields are:

- **High-Performance Computing (HPC):** Development of supercomputers for research and scientific work
- **Software Development:** Designing and developing software solutions for various industries
- **Cyber Security:** Working on data protection, security systems, and ethical hacking
- **Artificial Intelligence (AI) & Machine Learning (ML):** Developing smart systems and automation technologies
- **Embedded Systems & IoT:** Working on hardware-based programming and smart devices
- **Education and Training:** Providing various training programs for students and professionals

C-DAC is also known for conducting short-term and long-term courses that help students improve their practical skills and prepare for the IT industry.

Role in Education and Training

One of the important roles of C-DAC is to provide quality training to students. It organizes different training programs like programming, software development, and new technologies.

During my training, I attended a short-term course on “**OOP’s Concepts using Core Java**”. The training included both theory and practical sessions, which helped me understand the concepts clearly.

The training programs at C-DAC focus on real-world applications, which makes students industry-ready.

Organizational Structure

C-DAC follows a proper organizational structure to manage its work smoothly. The structure includes:

- **Director General:** Overall head of C-DAC
- **Centre Director:** Head of a specific center (like Patna)
- **Project Managers / Program Coordinators:** Manage projects and training programs
- **Faculty / Trainers:** Teach students and provide guidance
- **Technical Staff:** Help in lab work and system management

During the training, the sessions were conducted by **Mr. Vaibhav Patle** and **Ms. Nishu**, who guided us in both theory and practical work.



Fig. 1: Group Photo of Training Participants at C-DAC Patna